

Healthcare AI Notebooks

Executive Summary

This project consists of three end-to-end Jupyter notebooks that demonstrate practical skills in:

1. Large Language Model (LLM) fine-tuning using Unsloth and Gemma.
2. Multi-source data access using PostgreSQL, Google Drive, and Supabase.
3. Natural Language Processing (NLP) using DistilBERT for sentiment analysis.

The notebooks were designed to be technically accurate, visually polished, and production-inspired. Each notebook includes:

- Data preparation
 - Model training or data integration
 - Evaluation metrics
 - Visualizations
 - Interactive components
 - Final summary tables
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Repository Structure

```
healthcare-ai-notebooks/  
├── README.md  
├── requirements.txt  
  
├── 01_healthcare_llm_unsloth_gemma.ipynb  
├── 02_data_sources_postgres_drive_supabase.ipynb  
├── 03_bert_sentiment_analysis.ipynb  
  
├── assets/  
│   ├── notebook-1/  
│   ├── notebook-2/  
│   └── notebook-3/  
  
└── docs/  
    └── project_report.pdf
```

Notebook 1: Healthcare LLM with Unsloth and Gemma

Objective

To fine-tune Google's Gemma 2B Instruct model for healthcare question-answering using the Unsloth framework.

Technologies Used

- Unsloth
- Gemma 2 2B Instruct
- LoRA (Low-Rank Adaptation)
- 4-bit Quantization
- Hugging Face Transformers
- PyTorch
- Google Colab (Tesla T4 GPU)

Dataset

A synthetic healthcare dataset containing 300 question-answer pairs across ten symptom categories:

- Fever
- Cough
- Chest Pain
- Headache
- Stomach Issues
- Fatigue
- Diabetes Symptoms
- Skin Conditions
- Joint Pain
- Sleep Disorders

Training Configuration

Parameter	Value
Base Model	unsloth/gemma-2-2b-it-bnb-4bit
Fine-Tuning Method	LoRA
Dataset Size	300
Epochs	2
Trainable Parameters	19.6M
GPU	Tesla T4

Key Results

- Successful fine-tuning of Gemma for healthcare Q&A.
- Training loss decreased steadily.
- Interactive healthcare assistant implemented.
- Model generated relevant healthcare responses.

Example Output

Input: I have fever and chills.

Output: These symptoms may indicate a viral or bacterial infection. Rest, hydration, and consultation with a healthcare professional are recommended.

Visualizations

- Training loss curve
- Dataset preview

- Sample predictions
- Architecture diagram

Applications

- Symptom-based healthcare assistant
 - Telemedicine support chatbot
 - Patient education system
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Notebook 2: Multi-Source Data Access and Benchmarking

Objective

To demonstrate data retrieval and comparison from:

1. PostgreSQL (Neon)
2. Google Drive CSV
3. Supabase

Technologies Used

- Pandas
- SQLAlchemy
- psycopg2
- Supabase Python Client
- Neon PostgreSQL
- Google Drive

Dataset

Synthetic e-commerce dataset with:

- 100 records
- 11 features

Columns

- order_id
- customer_name
- product_name
- category
- quantity
- price
- order_date
- payment_method
- city
- order_status
- rating

Data Sources Compared

Source	Rows Retrieved	Columns	Storage Type
PostgreSQL (Neon)	10	11	Relational Database
Google Drive	100	11	CSV File Storage
Supabase	10	11	Backend-as-a-Service

Retrieval Time Benchmark

Source	Time (Seconds)
PostgreSQL	7.24
Google Drive	0.006
Supabase	0.77

Key Results

- Connected successfully to all three platforms.
- Verified schema consistency.
- Benchmarked retrieval performance.
- Generated comparison tables and charts.

Visualizations

- Rows retrieved chart
- Retrieval time chart
- Comparison summary table
- Architecture diagram

Applications

- ETL pipelines
- Cloud data integration
- Analytics workflows
- Dashboard backends

Notebook 3: BERT-Based Customer Review Sentiment Analysis

Objective

To fine-tune DistilBERT for binary sentiment classification of customer reviews.

Technologies Used

- DistilBERT

- Hugging Face Transformers
- PyTorch
- Scikit-learn
- Matplotlib

Dataset

Synthetic customer review dataset:

- 500 reviews
- Binary labels:
 - Positive (1)
 - Negative (0)

Training Configuration

Parameter	Value
Model	distilbert-base-uncased
Task	Sequence Classification
Epochs	2
Train/Test Split	80/20
Batch Size	16
Learning Rate	2e-5
Best Metric	F1 Score

Evaluation Results

Metric	Value
Accuracy	1.0000
Precision	1.0000
Recall	1.0000
F1 Score	1.0000
Eval Loss	0.3272

Confusion Matrix

Actual \ Predicted	Negative	Positive
Negative	47	0
Positive	0	53

Interactive Features

- Custom review prediction
- Confidence score display
- Probability bar chart

Example Predictions

Review	Prediction
This product is amazing and exceeded my expectations.	Positive
Very poor quality and not worth the money.	Negative
Delivery was quick and the packaging was excellent.	Positive

Visualizations

- Class distribution
- Training loss curve
- Confusion matrix
- Classification report
- Probability chart
- Workflow diagram

Business Applications

- Review analytics
- Customer satisfaction monitoring
- Product quality insights
- Brand reputation tracking

Comparative Summary

Notebook	Core Focus	Key Technologies
Notebook 1	LLM Fine-Tuning	Unsloth, Gemma, LoRA
Notebook 2	Data Engineering	PostgreSQL, Google Drive, Supabase
Notebook 3	NLP Classification	DistilBERT, Transformers

Skills Demonstrated

Machine Learning

- Transfer learning
- Fine-tuning transformer models
- Classification evaluation

Large Language Models

- LoRA fine-tuning
- Quantization
- Prompt formatting

Data Engineering

- PostgreSQL integration

- Supabase API usage
- Performance benchmarking

NLP

- Tokenization
- Sentiment analysis
- Interactive inference

Visualization

- Training curves
- Confusion matrix
- Comparative charts

Software Engineering

- Clean repository structure
 - Documentation
 - Reproducibility
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Key Learnings

- Fine-tuned a modern LLM for healthcare question-answering.
 - Integrated data from multiple cloud platforms.
 - Built an end-to-end BERT sentiment analysis pipeline.
 - Evaluated models with industry-standard metrics.
 - Created interactive notebook components.
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Limitations

- Synthetic datasets were used for demonstration purposes.
 - Real-world datasets would provide more robust validation.
 - Performance metrics may be optimistic due to dataset simplicity.
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Future Enhancements

Notebook 1

- Real medical datasets
- Retrieval-Augmented Generation (RAG)
- Multilingual support

Notebook 2

- Larger datasets
- Parallel data loading
- Additional warehouses (BigQuery, Snowflake)

Notebook 3

- Real Amazon/Yelp reviews
 - Multi-class sentiment analysis
 - Web deployment
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Conclusion

This project successfully demonstrates practical expertise in:

- Large Language Models (LLMs)
- Data Engineering
- Natural Language Processing
- Cloud Data Integration
- Model Evaluation and Visualization

All three notebooks are complete, reproducible, and organized in a professional repository structure suitable for internship submission and portfolio presentation.

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